

INSTITUTE OF ACTUARIES OF INDIA
EXAMINATIONS

May 2026

CP1B - Actuarial Practice

Time allowed: 3 Hours 15 Minutes

Total Marks: 100

Q.1) A country had implemented a Risk-Based Capital (RBC) regime for insurers, a couple of years ago. The adoption matched trends in neighbouring countries and ensured continued relevance in local and global insurance markets. After two years, the regulator observes the following:

- Insurers are holding capital comfortably above the minimum capital requirement.
- Product innovation has reduced significantly.
- Smaller insurers are increasingly exiting the market.
- Insurers' investment portfolios have shifted materially towards lower-volatility assets.
- Reinsurance arrangements have become more complex and concentrated with a few global reinsurers.

The regulator is concerned that the RBC regime may be having unintended consequences.

- i) What are the likely reasons for insurers to hold capital significantly above the regulatory minimum? [8]
 - ii) Explain how the introduction of RBC could have led to:
 - a) reduced product innovation [3]
 - b) market concentration [3]
 - c) changes in investment strategy [3]
 - d) increased reliance on reinsurance [3]
 - iii) Discuss whether these outcomes represent regulatory success or regulatory failure with respect to life insurance. [8]
 - iv) Outline potential refinements that the regulator could consider to address these concerns without weakening policyholder protection. [17]
- [45 marks]**

Q.2) A general insurer specialising in motor and household insurance has significantly expanded its use of cutting-edge methods in insurance (including but not limited to AI). They

- Use telematics and smart-home data
- Incorporate alternative data sources such as mobile phone usage and purchasing behaviour
- Continuously recalibrate pricing and underwriting decisions
- Are partially opaque ('black-box') in structure

Following implementation:

- Claims ratios have reduced.
 - Business mix has shifted towards digitally active policyholders.
 - Market share has increased significantly.
 - Certain demographic groups appear under-represented in new business.
 - Competitors are rapidly adopting similar AI models.
- i) How is the increased use of emerging technologies in underwriting likely to affect the insurer's risk profile in the short and long term? [8]

- ii) The regulator is concerned that widespread adoption of similar tech-based underwriting models across the market may:
- Increase similarity in risk selection and pricing approaches
 - Increased similarity in portfolios across insurers
 - Create correlated exposures and common technological dependencies

Discuss whether widespread adoption of these cutting-edge technologies could unintentionally increase correlation and vulnerability across insurers, potentially leading to sector-wide instability, even if individual firms experience improved risk selection.

Your answer should consider both short-term and longer-term effects. [10]

- iii) Certain demographic groups appear persistently under-represented in the insurer's AI-selected portfolio.

How would this move affect the long-term sustainability of the risk pool? Discuss potential implications for fairness and financial inclusion. [12]

- iv) Outline governance and risk management actions that the board should consider in response to the above developments. [6]

- v) The country where the insurer operates has a large population of young people employed in the services sector. With AI adoption increasing, there are fears of large-scale job losses in some parts of this sector. The insurer is considering launching a product that provides a financial benefit to individuals who are laid off due to AI-related workforce reductions. The product would pay a benefit until re-employment or for three months, whichever is sooner.

Assess the viability of this product. [15]

- vi) What changes would you recommend to make this more viable and have a stabler claims experience for the insurer. [4]

[55 marks]
